InShot

InShot is a powerful all-in-one Video and Photo Editor, as well as Video Maker phone application available both on Android and Apple devices. In this Software Critique paper i will specifically focus on the video editing interface of the software, however the image editing interface is very similar.

1. List the primary **goal(s)** of the users of the software system. Briefly describe who you feel are the **target users** of the system.
   The primary goals of users using the system is editing videos on the fly, when no laptop or PC is available. Editing encompasses everything from resizing/cropping the videos, stitching multiple videos together, overlaying videos with custom audio, adding effects, rotating the videos, adjusting video playback speed etc. I believe that the target users of the system are social media influencers, as I have quite often seen an InShot watermark on Instagram posts. I think it is designed to be popular with influencers, because their posts are often short and do not require too much editing time. InShot allows them to do these small edits on their mobile phone (which is also the common recording device). This convenience of editing videos on the recording device without the need of transferring them to a laptop or PC is what makes InShot so popular.

2. List and describe **three tasks** that it allows the user to perform, in order of importance; specify which of these would be considered the primary task.

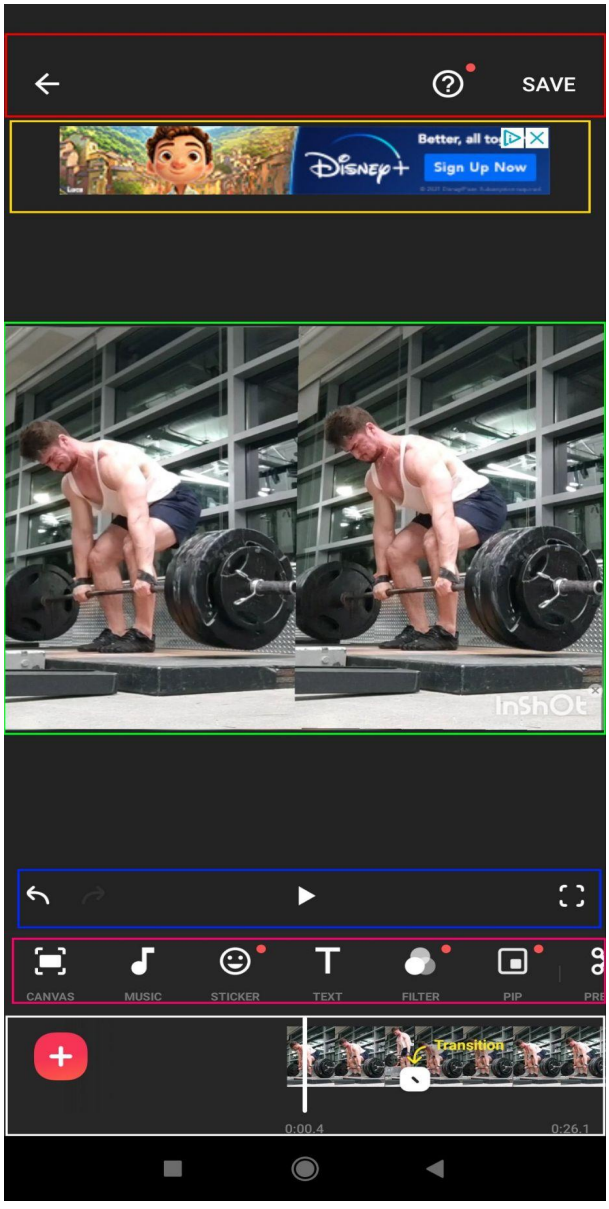
   1) Cropping/resizing videos
      Although video editing consists of a very complex set of actions, one of the most basic actions that almost any video edit utilizes is the cropping and resizing of said media. Often times the default camera application on a smartphone does not come with such an option. The videos are also often uploaded to social media sites which usually do not offer such editing capabilities in a graceful and easy-to-use manner as InShot does.

   2) Stitching many videos together
      The option of stitching the beginning of one video, to the end of another video, creating a single video consisting of the two initial videos played sequentially is also a common action when it comes to video editing. Default smartphone camera applications and social media sites often do not offer this capability.

   3) Adding music/custom sounds to videos
      Although some social media sites (e.g. Instagram) offer the option of adding music to a video, often times the selection of music is limited to songs licensed to the site. InShot allows the user to overlay any sound file over their video, bypassing the said restrictions of social media sites.

3. List **three** aspects of the interface that you feel are well designed, and **one** that you feel is not well designed (more if you feel appropriate or applicable), and describe why each feature is good or bad. Back up each choice using the design theories, principles, or guidelines discussed in class (reference Schneiderman's rules or Bertini's heuristics). For the aspect you feel is not well designed, suggest an alternative. Note that I'm <u>not</u> looking for likes and dislikes here; those are subjective measures.

   One aspect the interface does very well is its minimalist design and effective utilization of screen real estate, which allows for good screen readability (Bertini's Fourth and Fifth Heuristics). Let's look at an example from my most recent edit in InShot; where I combined two videos from one of my training sessions:



header area

advertisement area

image/video view

edit control area

toolbox area

rewind/fast-forward area

Figure 1. Effective use of screen real estate by InShot

The main area (marked in green in Figure 1) of the interface is the actual video we are editing. It takes the largest amount of space in the display, which makes sense since the results of the editing is the main thing the users will be focused on. All other areas take up a significantly smaller portion of the screen. The header area (marked in red in Figure 1) contains a few action buttons for finalizing the video; a button for exiting out of the project; and a button for displaying more information. The advertisement section (marked in yellow in Figure 1) is big enough so that the user can see and read what the advertisement is about, but also not too large to impede the user from using the application. The edit control area (marked in blue in Figure 1) allows the user to pause/play a video, to undo/redo his actions and to maximize the video view. The toolbox area (marked in pink in Figure 1) neatly displays all the tools available to the user. Finally the rewind/fast-forward area (marked in white in Figure 1) allows the user to easily navigate to the exact frame in the edited video to check the results of their edit.

One other good aspect of the interface is that it provides informative feedback and dialogue when it comes to user actions (Bertini's Third Heuristic and

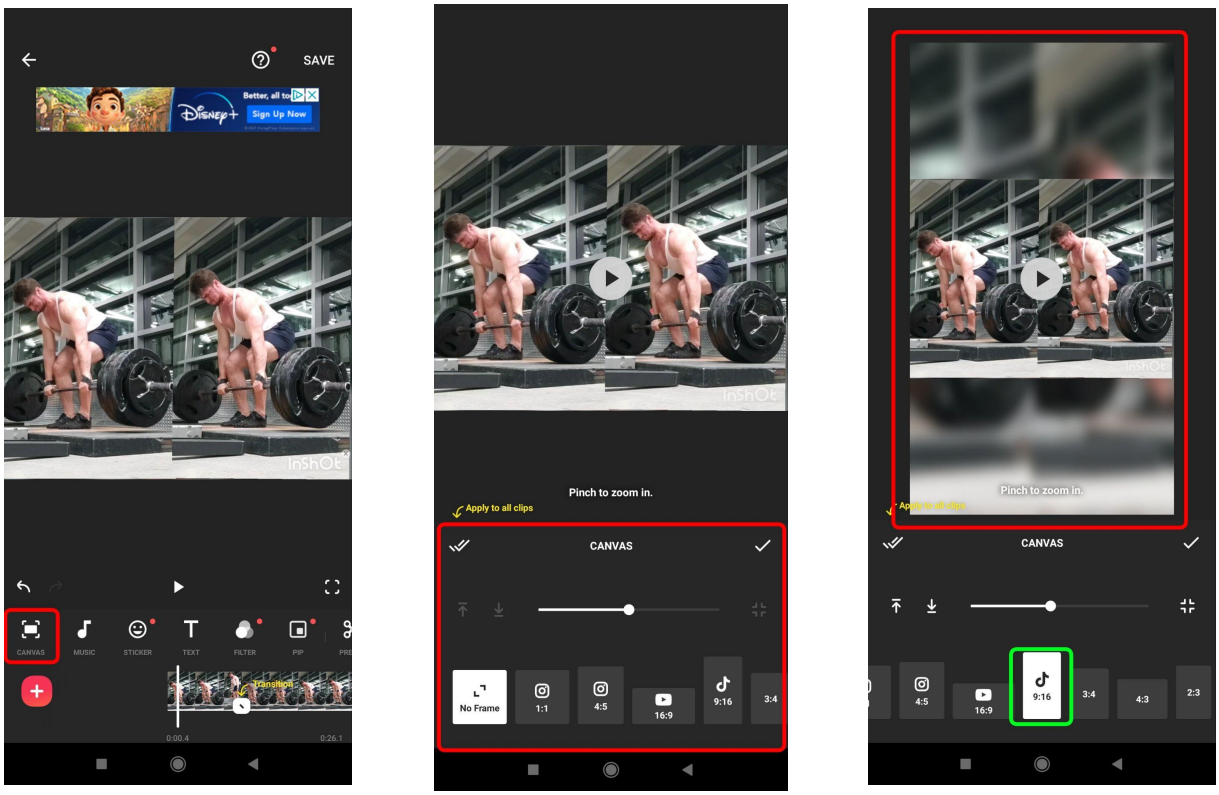


Figure 2. Effective context switching when using a particular tool, and visual feedback of what the final result of using such tool looks like

Schneiderman's Rules Three and Four). For example, when selecting a tool there will be a context switch, indicating that a tool selection has been made. Suppose the user wishes to change the canvas size. He selects the "canvas" tool indicated in red in the left image of Figure 2. Once selected, there will immediately be a context switch as seen in the bottom of the screen indicated in red in the center image of Figure 2. He no longer sees the full toolbox, but rather the specific options the canvas tool offers. Now the user knows that his tool selection was successful. Once he makes a particular selection (e.g. a specific canvas aspect ratio), that specific ratio is highlighted as shown in green in the right image of Figure 2, and there is immediate informative visual feedback (indicated in red in the right image of Figure 2), as to how applying this action affects the video.

Another good example of effective dialogue utilization is when finalizing a video project. Clicking on the save button in the top right brings the user to a dialogue screen where he can indicate what quality and frame rate he wants the finalized video to have, and the interface responds with an estimate of how much storage space this finalized video will have (left image of Figure 3).

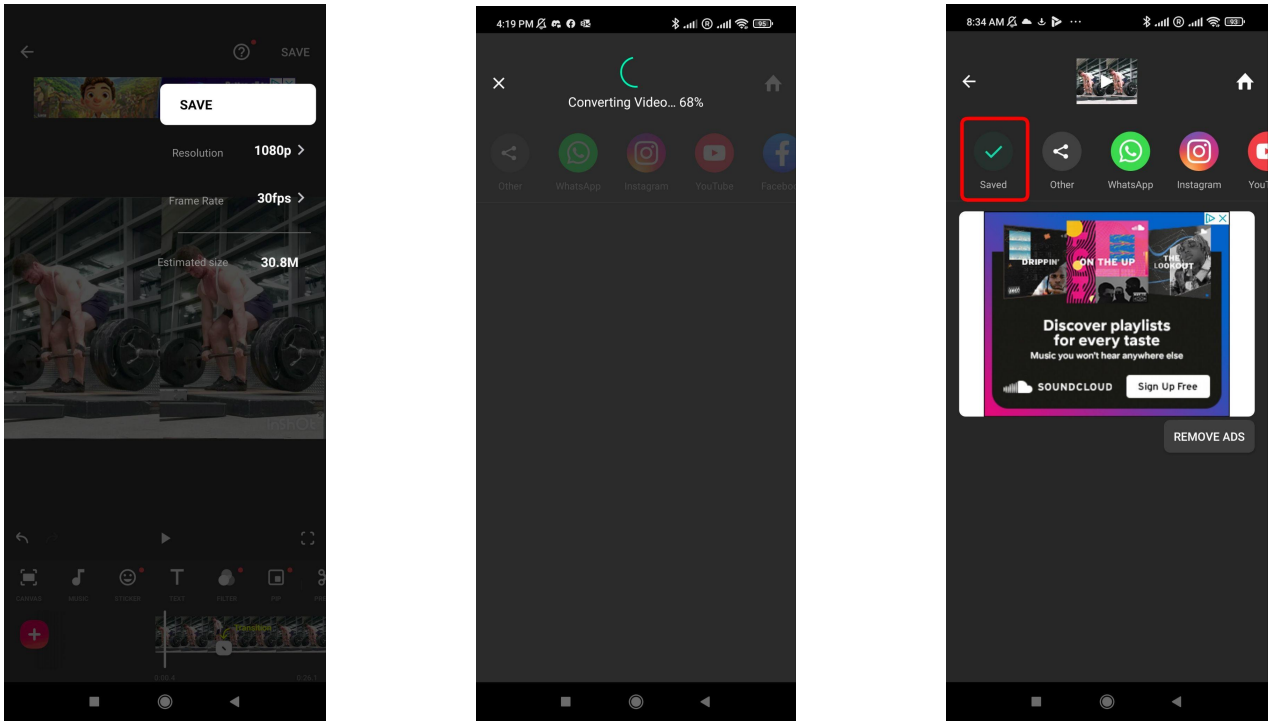


Figure 3. Effective use of dialogue and informative feedback when finalizing a video project

Once the selection of quality and frame rate is made, the user is brought to another screen where the interface assures them in the form of dialogue at the top of the screen (center image of Figure 3) that their video is being finalized and how far it is along the process. Once the app finishes, the user is shown a small

thumbnail of their video in the top of the interface, where previously the dialogue box was placed, and a small icon saying "Saved" indicating completion, as indicated in red of the right image of Figure 3. This could also be thought of as an application of Bertini's first heuristic, i.e. visibility of system status.

The third aspect the interface does well is allowing users to undo actions (Schneiderman's Rule Six) as well as keeping users in control at all times (Schneiderman's Rule Seven). If we look back to Figure 1, we can see that inside the edit control area marked in blue, on the left we have undo and re-do buttons. These are extremely important when it comes to video editing, as one usually makes many mistakes. InShot also always keeps users in control, by allowing them to return to the home page at any point (with icon indicated in red on the left image of Figure 4, and icon indicated in green on the right image of Figure 4). Once the user does so, their video project is saved as a draft. When the user is finalizing the video, a potentially time consuming action, the app allows them to cancel this action with the icon indicated in red on the center image of Figure 4. Once the app is done finalizing the video, it allows the user to return to editing the video with the icon indicated in red on the right image of Figure 4.

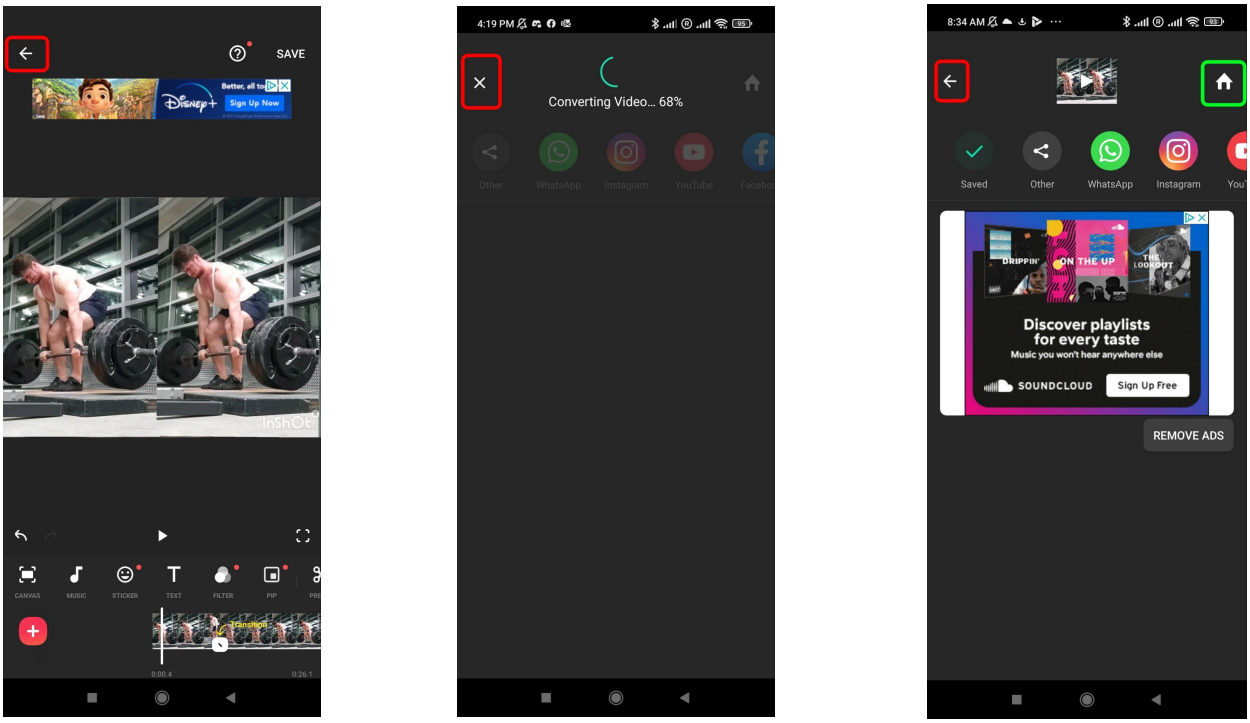


Figure 4. Icons which allow the user to always be in control i.e. to exit the editing process at their own will, or cancel a time consuming action at any point

The thing that InShot (or at least the free version of it) does not really support is customization (Berini's Sixth Heuristic and Schneiderman's Second Rule). For example, a user might use a select few tools more commonly than others.

Suppose a set of tools indicated in red of Figure 5 is used quite frequently by a user.

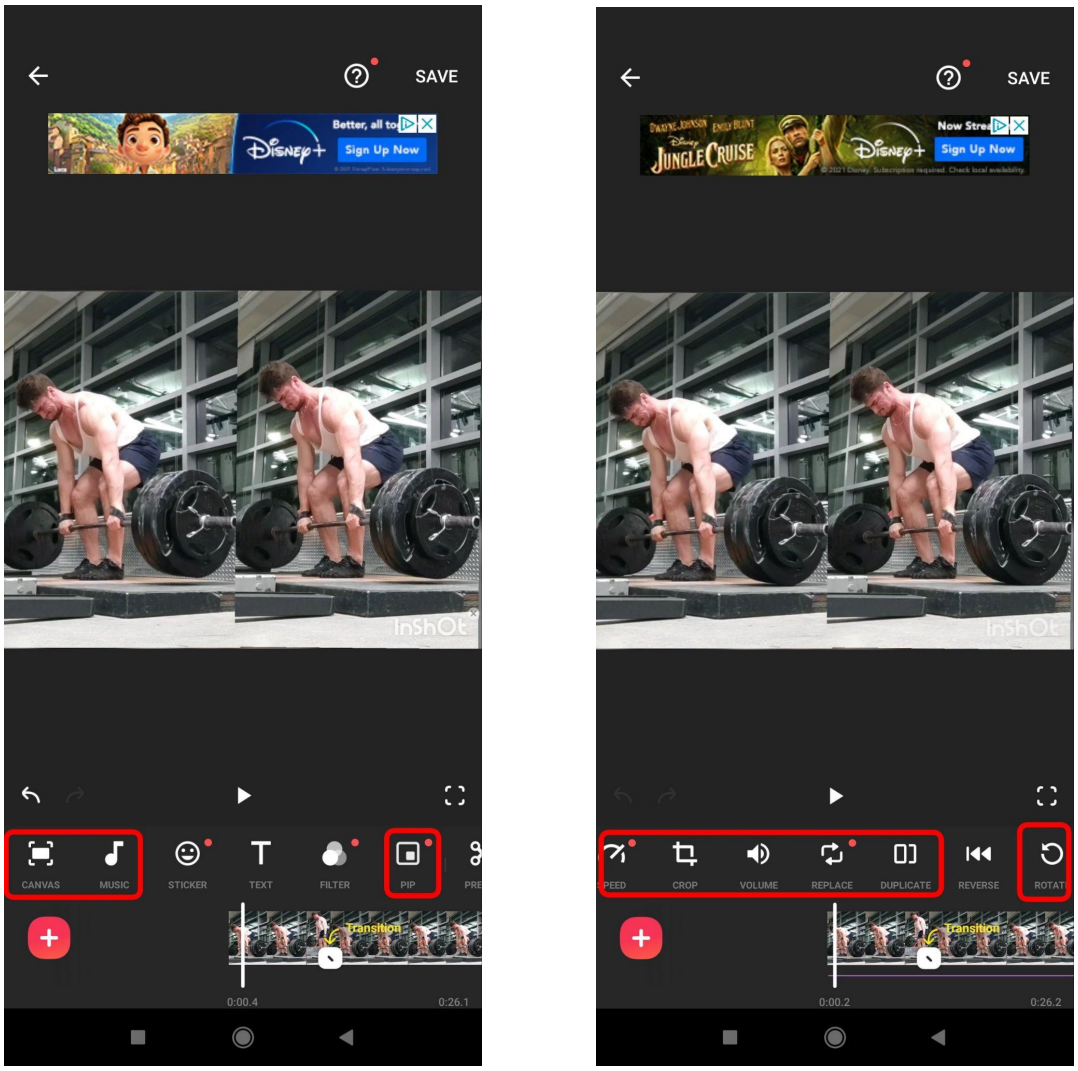


Figure 5. A possible set of tools which are used commonly by an individual user

However, to get to those tools the user has to scroll past, and in his mind filter out all of the tools he most likely will not use. It would be much better if InShot allowed the user to customize the layout of the toolbox, allowing him to, for example, place the most commonly used tools on the left hand side of the toolbox, which would allow a faster tool selection.

4. Analyze the one of major functions it performs and derive a requirements specification for that one task, i.e., a document describing *what* the software was designed to do, but not *how* it does it. Then suggest at least two alternative designs for part of the interface based on this requirements specification. Again, back up your designs with theories, principles, or guidelines.

Here I will analyze the function of changing the canvas aspect ratio. The canvas aspect ratio is the aspect ratio of the finalized video (the ratio of its width divided by its height). It determines the frame inside which all video segments must fit. For example, suppose one wants to stitch multiple videos together, each having a different aspect ratio. The canvas aspect ratio determines what the aspect ratio of the resulting video will be; forcing each video segment to be either resized or cropped in order to fit in this frame.

One tool of the interface called "Canvas" is designed to do just this. It allows the user to change the current aspect ratio (e.g. 4:3) to a new aspect ratio (e.g. 1:1).

Currently how the app implements this functionality is already shown in Figure 2. The user can select the aspect ratio from a range of pre-specified aspect ratios. The approach works well for most purposes. However there are two obvious drawback: custom aspect ratios are not allowed, and finding the desired one might be time consuming.

An alternative approach to the current one would be to offer the user the ability to select an aspect ratio from a range of "reasonable" aspect ratios (say from 1:8 to 8:1) using a slider marked in green in Figure 6.

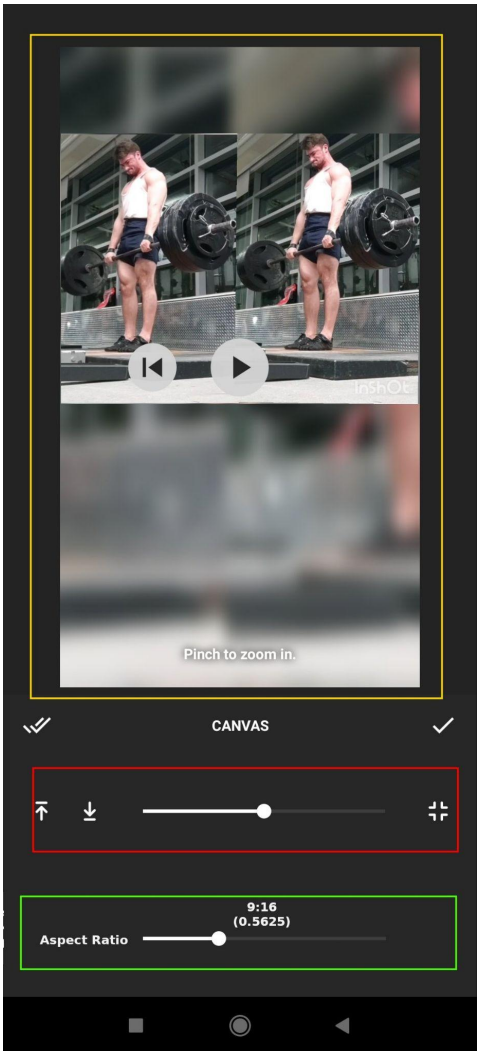


Figure 6. First alternative for the implementation of changing the Canvas aspect ratio using a slider.

The slider and buttons indicated in red of Figrue 6 are the same as in the original implementation. They allow the user to scale the image segment relative to the canvas and place it accordingly. The slider marked in green is the added slider allowing for smooth aspect ratio changes. Such implementation for aspect ratio changes follows Bertini's Fourth Heuristic which advocates for minimalist design. This type of design also decreases the cognitive workload of the user (Schneiderman's Eight Rule) and facilitates ease of input (Bertini's Fifth Heuristic), by instead of the user having to look through a vast number of different aspect ratios to find the desired one, he has one simple slider which allows him to quickly and easily "fine tune" it. The slider also comes with a text dialogue, which shows what the numerical value of the aspect ratio is (e.g. 0.5625), as well as a fractional representation (e.g. 9:16) (Schneiderman's Fourth Heuristic). Moving the slider would cause the canvas to continuously transform in the video view marked in yellow, similarly to the original implementation. This gives users informative feedback (Bertini's Third Heuristic and Schneiderman's Third Rule) showing the direct impact of their actions on the final video product.

A second alternative approach would be to adjust the aspect ratio using an interactive grid overlay indicated in red in Figure 7.

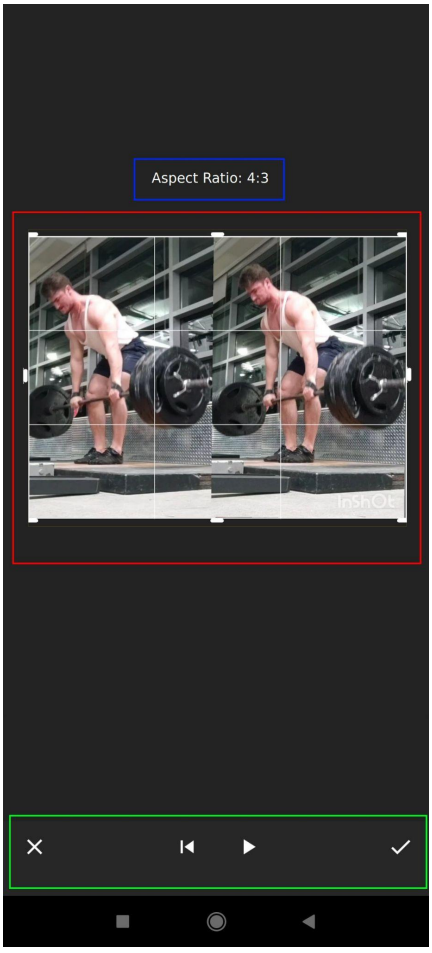


Figure 7. Second alternative for the implementation of changing the Canvas aspect ratio using an interactive grid overlay.

The user adjusts the aspect ratio by placing their finger at one of the eight bolded boxes on the corners or side bisectors of the gird overlay, and simply dragging them into the desired shape. The box indicated in green contains video pause/play controls, as well as buttons for applying or cancelling the changes made. Just like the previous alternative, this approach follows Bertini's Fourth Heuristic which advocates for minimalist design, where instead of having many aspect ratios to choose from, now only a single box overlay is used for resizing. For the same reasons as the previous alternative, this design also decreases the cognitive workload of the user (Schneiderman's Eight Rule) and facilitates ease of input (Bertini's Fifth Heuristic). A text dialogue (indicated in blue), is placed directly above the canvas view and shows a fractional representation (e.g. 4:3) of the aspect ratio, satisfying Schneiderman's Fourth Rule (dialogue for closure). By placing the dialogue box directly above the canvas view screen readability and glanceability are also improved(Bertini's Fifth Heuristic), since the user's gaze does not have to wander too far from the grid overlay they are resizing, when they wish to check what the current aspect ratio is. Resizing the box overlay would cause the canvas to continuously transform in the canvas view similarly to the previous alternative and original implementation. This gives users informative feedback (Bertini's Third Heuristic and Schneiderman's Third Rule) showing the direct impact of their actions on the final video product. This alternative also offers some consistency (Bertini's Third Heuristic and Schneiderman's Firtst Rule), as the application already implements a similar interface for cropping individual video segments.